Energy Causes Change

Chapter 1 Guided Notes · Grade 4 Energy Transfer & Transformation

Name: _____ **Date:** _____

BIG QUESTION

Where can we observe evidence of energy causing change?

KEY WORDS

energy — Energy is the ability to cause _____ .

change — Change means something becomes _____ .

evidence — Evidence is something we observe that helps us _____ what happened.

THE MODEL

ENERGY → **CHANGE** → **EVIDENCE**

In my own words: energy causes a change, and the change is our evidence.

ONE EXAMPLE — BEFORE & AFTER

Pick one change from class. Draw what it looked like before and after.

BEFORE	AFTER

What changed? _____

FORMS OF ENERGY

MOTION	LIGHT	SOUND	HEAT	ELECTRICAL
rolling	*bright*	*loud*	*warm*	*powers*

Stored Energy

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Name: _____ **Date:** _____

KEY IDEA

Stored energy can cause a change later.

THREE EXAMPLES

For each, write the change that could happen later.

battery

in a flashlight

What change could happen later?

_____

stretched rubber band

held still

What change could happen later?

_____

raised object

held up high

What change could happen later?

_____

OBSERVATION VS EXPLANATION

Observation = what I see. Explanation = what it means.

Example: "The light turned on" is an observation. "Energy caused the change" is an explanation.

DRAW ONE STORED-ENERGY EXAMPLE

draw your stored-energy example here

Energy Detectives — Evidence

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Name: _____ **Date:** _____

EVIDENCE TABLE

Record one change you found as an Energy Detective.

Before _____

After _____

What changed? _____

What is the evidence? _____

SYNTHESIS

How do scientists know energy is involved if energy itself cannot be seen directly?

REMEMBER

ENERGY	CHANGE	EVIDENCE